

SEO Optimization Agents:

Video SEO Optimizer: Project Overview :

Purpose

This is a web application that analyzes video content (primarily YouTube videos) and generates platform-specific SEO recommendations to help content creators optimize their videos for better discoverability and engagement.

Video SEO Optimizer PRO

 Analyzes YouTube videos and creates AI-powered SEO optimizations

 Uses OpenAI GPT-4 to generate tags, descriptions, and timestamps

 Creates thumbnail concepts with DALL-E integration

 Deploy using Docker, GitHub Actions, and AWS EC2/ECR

Tools & Technologies

-  Python
-  Streamlit (web interface)
-  OpenAI GPT-4 API (content analysis)
-  DALL-E 3 (thumbnail generation)
-  Google Gemini API (alternative AI model)
-  CrewAI (agent framework)

Components

-  User Interface (Streamlit)
-  Video Metadata Extractor
-  Content Analysis Engine
-  SEO Recommendation Generator
-  Thumbnail Designer
-  Multi-language Support

Architecture

- Frontend: Streamlit web application
- Backend: Python modules for analysis and API interactions
- AI Services: OpenAI GPT-4, DALL-E 3, and Google Gemini

Workflow

1. User inputs a video URL and selects output language
2. App extracts video metadata (title, author, duration, etc.)
3. AI analyzes video content based on metadata
4. App generates SEO recommendations:
 - 35 trending tags/hashtags
 - Optimized video description
 - Strategic timestamps
 - Alternative title suggestions
 - Thumbnail concepts

```
video-seo-optimizer/  
├─ app.py           # Main application file  
├─ requirements.txt # Dependencies  
├─ utils/  
│   ├── __init__.py  
│   ├── video_extractor.py # Extract video metadata  
│   ├── analysis_functions.py # Analysis functions  
│   └─ thumbnails.py      # Thumbnail generation  
├─ assets/          # Static assets  
└─ README.md        # Documentation
```

Video SEO Optimizer Pro



API Configuration

OpenAI API Key

Google Gemini API Key

Language Settings

English ▼

About

This tool uses AI to analyze videos and generate SEO recommendations

Enter video URL

https://www.youtube.com/watch?v=...

Analyze

Video Details

YouTube

Title: How to Optimize YouTube Videos for SEO

Creator: SEO Expert • Duration: 10m 30s • Views: 15,42

Current Thumbnail

Analysis

Tags (35)

Description

Timestamps

Titles

Thumbnails

Content Analysis

This video provides a detailed guide on optimizing YouTube videos for better search engine rankings. The content is structured as a step-by-step tutorial with practical examples and demonstrations of effective techniques.

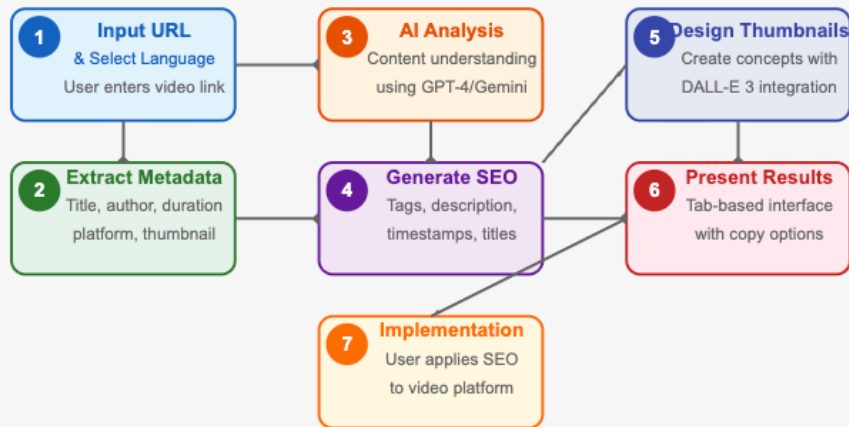
Main topics covered:

- Keyword research for video titles and descriptions
- Creating compelling thumbnails that drive clicks
- Using tags effectively to improve discoverability

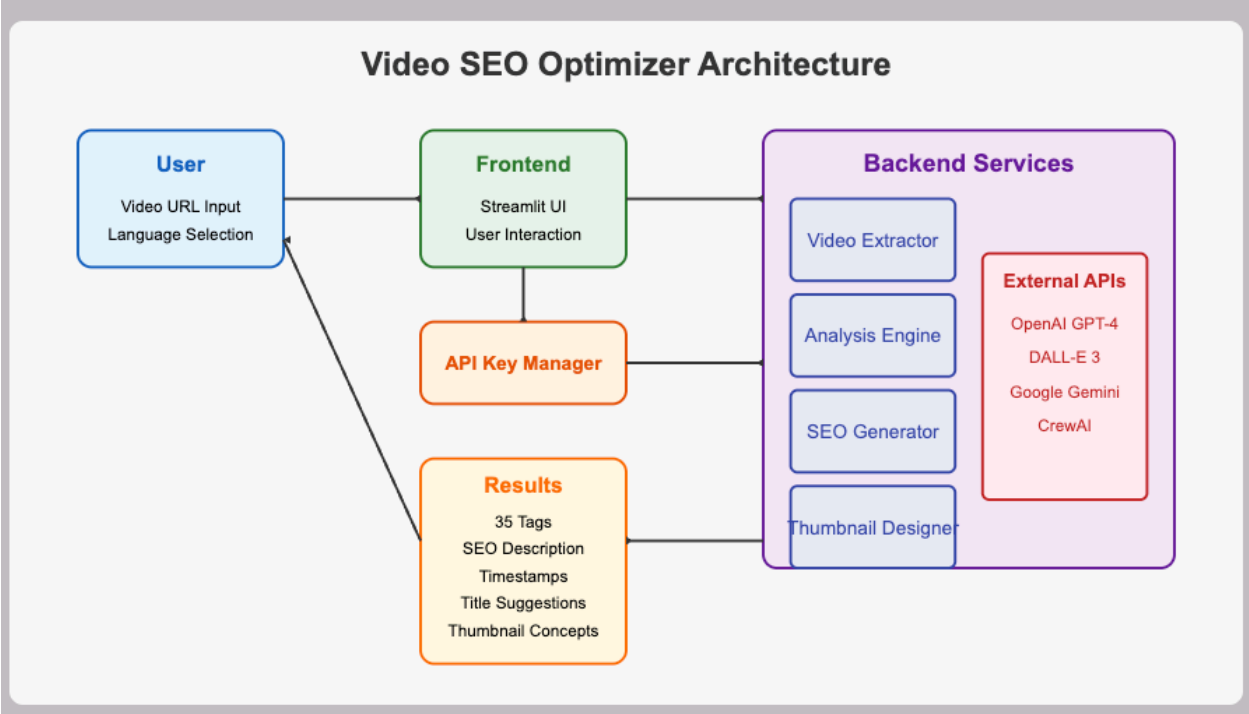
SEO Optimization Agents:

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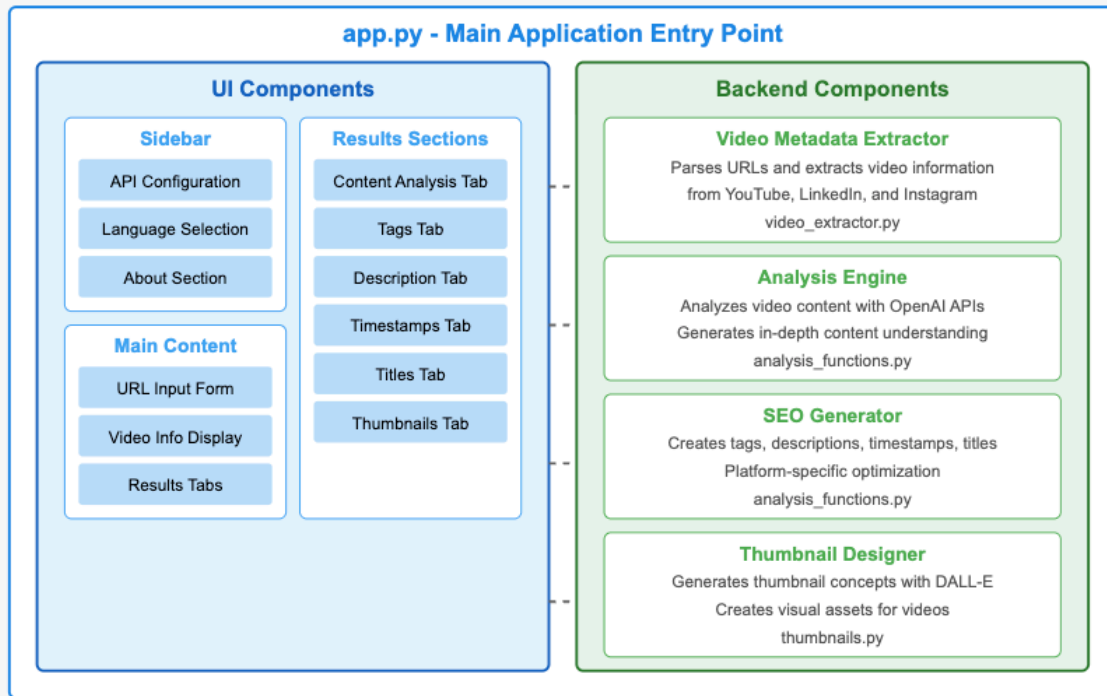
Video SEO Optimizer Workflow



Multi-language Support: English, Spanish, French, German, Italian, Portuguese, Hindi, Japanese, Korean, Chinese, Russian, Arabic

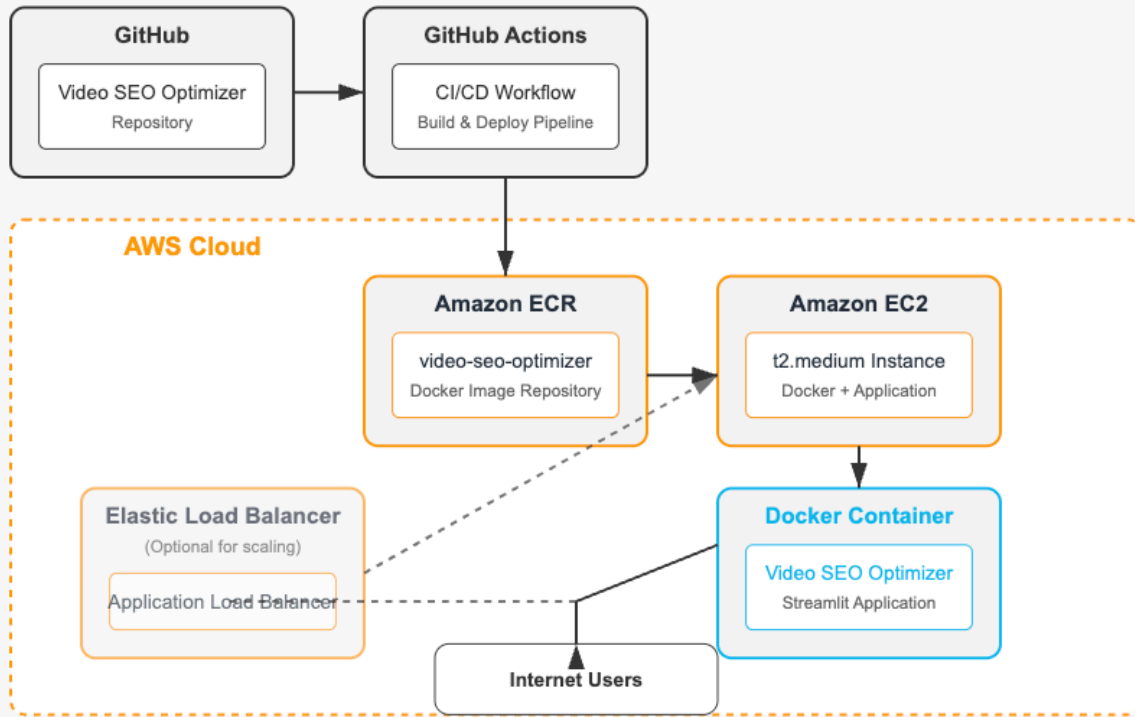


Video SEO Optimizer Component Structure



Deployment Architecture :

AWS Deployment Architecture for Video SEO Optimizer



AWS Deployment Flow for Video SEO Optimizer

